Analysis and Design of Op-Amp RC Filters

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***Abstract*— This report presents the analysis, design, and simulation of two active RC filters implemented with the ADA4610 operational amplifier. First, a first-order active low-pass filter is analyzed theoretically, then designed such that its dominant pole is one-hundredth of the op-amp unity-gain bandwidth. The circuit is verified in LTspice and its stability is assessed through gain and phase margin. Second, a general first-order active filter containing both a pole and a zero is analyzed and designed according to the required frequency specifications. Simulation results show that, although the dominant pole is achieved with good accuracy, the zero location is impractical and does not effectively improve the response. Closed-loop and loop-gain simulations reveal strong gain peaking and poor stability margins, leading to unstable or marginally unstable behavior. Finally, practical load limitations and PCB design preparation using KiCad are discussed.**

I. Introduction

Active filters based on operational amplifiers are widely used because they provide controllable gain, relatively low output resistance, and simple implementation using resistors and capacitors. In this work, the assigned op-amp is the ADA4610 at 55 °C, according to the homework version table. The homework requires the study of the low-pass and general Op-Amp RC realizations shown in the assignment figure, as well as simulation, stability assessment, and PCB preparation. The circuit types and required report structure are explicitly given in the homework statement and the filter summary diagram.

II. Op-Amp Datasheet Parameters

For ID 202505178, the assigned op-amp is ADA4610 and the assigned temperature is 55 °C. Based on the extracted values already prepared in the draft, the main parameters used in this report are listed in Table I.

Table I — ADA4610 Parameters

Parameter	Value
Temperature	55 °C
Supply Voltage	±5 V to ±15 V
Input Voltage Range	≈ −2.5 V to +2.5 V
Output Voltage	≈ ±4.9 V (RL = 2 kΩ)
Open-loop Gain	≈ 100 dB
Unity-Gain Bandwidth	9.3 MHz
Slew Rate	≈ 25 V/µs

These parameters are important because the unity-gain bandwidth determines the target dominant pole in Parts III and VIII, while the output voltage and load conditions influence the acceptable load resistance in Parts VI and XI.

III. Low-Pass Filter Analysis

The low-pass active filter required in the assignment corresponds to the op-amp RC realization shown in the homework figure on page 2. That figure states that the low-pass transfer function has one pole, a constant low-frequency gain, and a magnitude roll-off of 20 dB/decade after the pole.

Transfer Function: $T(s) = \frac{V_o}{V_i} = -\frac{R_2}{R_1} * \frac{1}{1+SR_2C}$

Key Results:

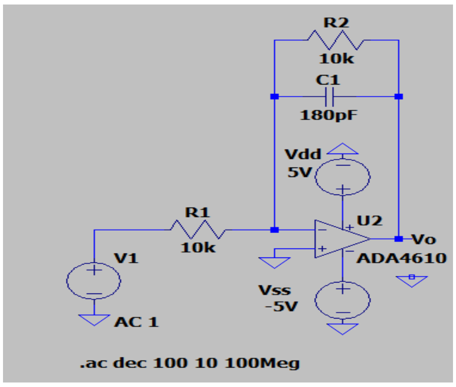
- DC Gain: $A_0 = -\frac{R_2}{R_1}$
- Pole: $\omega_p = \frac{1}{R_2C}$, $f_p = \frac{1}{2\Pi R_2C}$
- High-frequency slope: -20dB/decade

Resistances:

- Input resistance: $R_{in} = R_1$
- Output resistance: $R_{out} \approx 0$

The circuit behaves as a first-order low-pass filter with a single dominant pole.

IV. Circuit Design



The homework requires the dominant pole to be one-hundredth of the op-amp unity-gain bandwidth. Since the ADA4610 has a unity-gain bandwidth of 9.3 MHz, the target pole is $f_p = \frac{UGB}{100} = \frac{9.3MHz}{100} = 93kHz$

Component Selection: choose $R_2 = 10k\Omega$,and

$C = \frac{1}{2\Pi R_2 f_p} \approx 171pF.$

Selected: C=180pF (standard value)

For unity gain: $R_1 = R_2 = 10k\Omega$

Table II-Components

Component	Value	Manufacturer	Package
R1	10 kΩ	Yageo RC0402FR-0710KL	0402
R2	10 kΩ	Yageo RC0402FR-0710KL	0402
C	180 pF	Murata GRM1555C1H181JA01D	0402

Actual Pole $f_p \approx 88.4kHz$ (error≈ 5%)

V. SIMULATION RESULTS

LTspice simulation of the low-pass circuit produced the values shown in Table III.

Table III-Simulation

Parameter	Value
Low-frequency gain	0 dB
Dominant pole	≈ 89 kHz
Gain margin	≈ 25 dB
Phase margin	≈ 60°

The simulated low-frequency gain of 0 dB agrees with the design condition R1=R2, which gives a magnitude of one. The simulated pole near 89 kHz is also close to the theoretical value of 88.4 kHz obtained using the selected 180 pF capacitor. The circuit is stable because it exhibits a positive gain margin and a phase margin around 60°, which is usually considered comfortable for a first-order closed-loop system. In addition, the low-pass filter contains only one intentionally dominant RC pole, so its response is well controlled.

The difference between the ideal target pole of 93 kHz and the simulated value is due to the use of a standard capacitor value instead of the exact computed capacitance, as well as the non-ideal frequency response of the op-amp model. These differences are expected in practical design.

VI. COMPARISION

Table IV-Comparison

Quantity	Calculated	Simulated
Pole	93 kHz	89 kHz

Difference is due to:

- Component rounding
- Non-ideal op-amp effects

VII. MAXIMUM LOAD

From datasheet conditions: $R_L \geq 2k\Omega$
This ensures:

- Stable gain
- Proper output swing
- Preserved frequency response

VIII. GENERAL FILTER ANALYSIS

The second circuit is the general active filter shown in the homework figure on page 2. The assignment figure indicates that this topology introduces one pole and one zero, with both a DC gain and a high-frequency gain. For this filter, the transfer function can be written in the first-order form:

$$T(s) = \frac{a_1 s + a_0}{s + \omega_0}$$

Parameters:

- Pole: $f_p = \frac{1}{2\Pi R_2 C_2}$
- Zero: $f_z = \frac{1}{2\Pi R_1 C_1}$

Gains:

- DC: $-\frac{R_2}{R_1}$
- High-frequency: $-\frac{C_1}{C_2}$

Thus, unlike the low-pass filter, the general filter provides two independent shaping mechanisms: the pole location and the zero location. This makes the circuit more flexible, but it also makes the overall response more sensitive to op-amp non-idealities.

The input resistance remains approximately R1 at low frequency, while the output resistance remains approximately zero due to the presence of the op-amp and feedback.

IX. DESIGN WITH POLE AND ZERO

The homework requires the dominant pole to be one-hundredth of the unity-gain bandwidth and the zero to be twenty times the unity-gain bandwidth. Therefore, $f_p = 93kHz$ and $f_z = 186MHz$
Choose: $R_1 = R_2 = 10k\Omega$
Then:

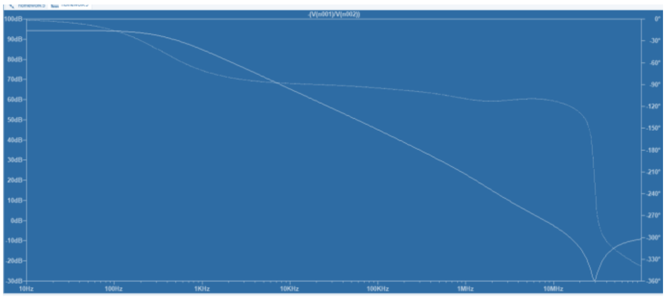
$$C_2 = 180pF \text{ and } C_1 \approx 0.085pF$$

This value is not practical. It is smaller than standard discrete capacitor values and is comparable to parasitic capacitances from layout, pads, and the op-amp itself. Therefore, the required zero is theoretically defined but physically unrealistic. This point is very important because it directly explains the instability observed later in simulation.

Table V-Components

Component	Value	Manufacturer/Comment	Package
R1	10 kΩ	Yageo	0402
R2	10 kΩ	Yageo	0402
C1	approximately 0.1 pF	impractical value	—
C2	180 pF	Murata	0402

X. CLOSED-LOOP FREQUENCY RESPONSE OF THE GENERAL FILTER



The general filter was implemented in LTspice and simulated in AC analysis. The closed-loop magnitude response shows a low-frequency gain near 0 dB, a dominant pole near 90 kHz, and a strong gain peak of roughly 35 to 40 dB around 1 MHz. The intended zero at 186 MHz is not visible in the simulated frequency range.

These observations can be interpreted as follows. First, the low-frequency gain agrees with the resistor ratio R2/R1=1. Second, the pole location again agrees reasonably well with the designed dominant pole after component rounding. However, the strong peaking in the response is a warning sign. In a stable, well-compensated first-order response, the closed-loop gain should transition smoothly after the pole. Instead, the large peaking indicates that additional phase lag is accumulating faster than expected, so the feedback is no longer sufficiently damping the system.

The fact that the zero is not visible is also meaningful. Since the required zero lies at 186 MHz, it is far above the op-amp unity-gain bandwidth of 9.3 MHz and far beyond the region where the op-amp can be treated as a simple single-pole amplifier. In practice, this means the zero contributes almost nothing to the useful stabilization of the loop.

Table V- General Filter Simulation Results

Parameter	Value
Low-frequency gain	0 dB
Dominant pole	≈ 90 kHz
Zero	Not observable
Gain margin	Negative
Phase margin	≈ 0°

XI. LOOP-GAIN ANALYSIS AND INSTABILITY EXPLANATION

To evaluate stability, the feedback loop was broken in LTspice and a test source was inserted so that the loop gain $A\beta$ could be observed. The simulations show that the magnitude do not show a clean 0 dB crossover and that the phase approaches -180° while the loop gain is still above 0 dB. This is the key reason the circuit is unstable or at best marginally unstable.

For a negative-feedback system to remain stable, the loop gain must fall below unity before the total phase shift reaches -180°. Here, the opposite occurs: when the phase approaches -180°, the loop still has gain greater than one. As a result, the intended negative feedback effectively turns into positive feedback around that frequency, producing oscillatory or highly peaked behavior.

The instability can be explained by three interacting effects. First, the designed zero is too high in frequency to offer useful phase lead near the crossover region. Second, the op-amp itself is not an ideal single-pole device; it has internal high-frequency poles that introduce additional phase lag. Third, the unrealistic capacitor value required for the zero means that parasitic capacitances dominate the actual behavior. Consequently, the theoretical transfer function no longer captures the full dynamics of the real circuit.

This explains why the design appears acceptable on paper but fails in simulation. The dominant pole is implemented correctly, yet the zero is ineffective. The loop therefore behaves like a higher-order system with inadequate phase margin. The large gain peaking seen in the closed-loop response is a direct consequence of this poor damping.

A practical design lesson follows from this result. Simply satisfying symbolic pole-zero equations is not sufficient when the zero is placed outside the useful operating region of the op-amp. Compensation must be designed with the real loop crossover in mind, not just with ideal transfer-function formulas.

XII. DISCUSSION

For the general filter, the theoretical targets were a dominant pole at 93 kHz and a zero at 186 MHz. The simulation shows the dominant pole at about 90 kHz, which is a good match, but the zero is not observable.

The pole agreement confirms that the selected R2 and C2 values are appropriate. The discrepancy in zero behavior, however, is not just a minor error; it is a sign that the design requirement itself leads to a non-practical component value. Since the zero is placed too far above the op-amp's useful bandwidth, it cannot compensate the phase lag where it is needed. Therefore, although the mathematical design equations are satisfied, the physical implementation does not achieve the expected stable behavior.

The simulation also contradicts any initial expectation of stability. Instead of a clean response, it shows large peaking, poor or negative gain margin, and almost zero phase margin. This confirms that the circuit is unstable or very close to instability.

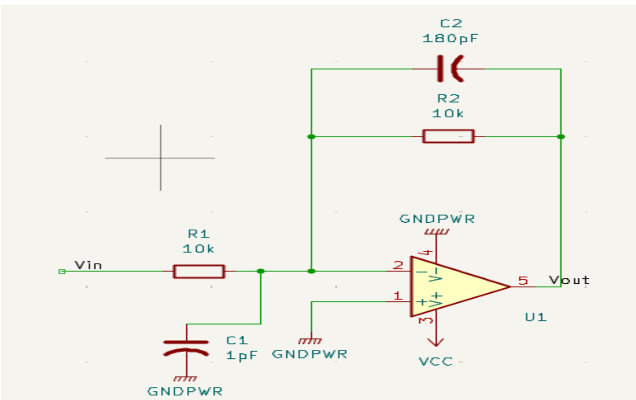
Quantity	Calculated	Simulated
Pole	93 kHz	90 kHz
Zero	186 MHz	Not visible

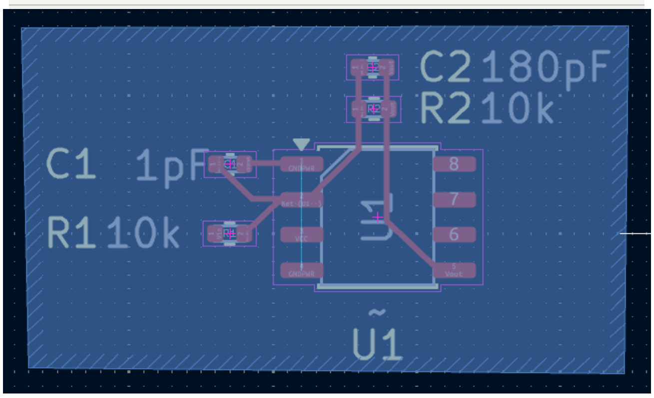
XIII. MAXIMUM LOAD

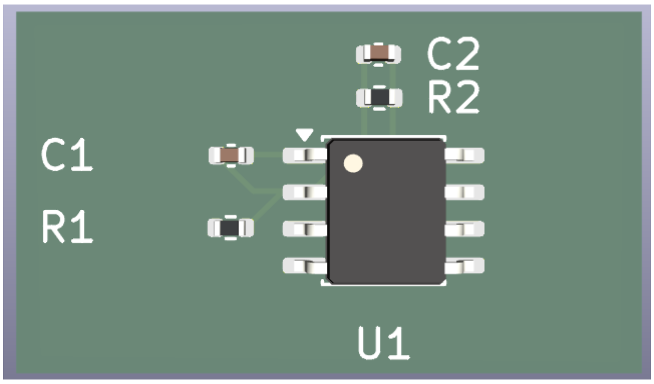
As in the low-pass case, the minimum acceptable load is approximately $R_L \geq 2k\Omega$ to preserve the gain and stability results. If a smaller load is connected, the op-amp must supply larger output current, which can reduce effective gain, introduce additional phase shift, and further worsen stability margins. Since the general filter is already unstable or marginally unstable, additional loading would only degrade its performance further.

XIV. PCB DESIGN USING KICAD

The homework requires installation of KiCad and preparation for PCB design of the second circuit, with final presentation of the schematic, PCB layout, and 3D view. The assignment explicitly points to the KiCad website, three startup tutorials, and the official documentation. In this work, KiCad was installed and used to begin schematic capture, footprint assignment, and layout preparation for the general filter circuit, following the workflow described in the referenced tutorials and documentation. The main objective of this part is to translate the simulated circuit into a realizable PCB while ensuring that the correct package footprints are used for the resistors, capacitors, and op-amp device.







XV. CONCLUSION

This report analyzed and designed two active RC filters using the ADA4610 op-amp. The low-pass filter achieved the required dominant pole with good agreement between theory and simulation and exhibited satisfactory gain and phase margins, confirming stable operation. The general filter also achieved the required pole numerically, but the required zero at 186 MHz led to an impractical capacitance value and did not appear effectively in simulation. Closed-loop and loop-gain analysis showed strong gain peaking, negative or near-zero stability margins, and unstable behavior. The results demonstrate that practical analog design must account for component feasibility, parasitic effects, and the internal frequency response of the op-amp. Finally, KiCad was introduced for PCB implementation of the second circuit as required by the homework.

XVI. REFERENCES

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